Intelligence Tipper

CSIT-23234

BreachedForums Record Analysis Reveals Value of Compromised Credentials for Data-Leak Advertisers and Purchasers

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BreachedForums Record Analysis Reveals Value of Compromised Credentials for Data-Leak Advertisers and Purchasers

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Key Points

- Despite the brief tenure of BreachedForums (a.k.a. Breached), the English-language eCrime forum's records and member behavior provides insight into trends within the eCrime ecosystem. Many Breached members were almost certainly previous RaidForums members, and their forum activity illustrates the supply and demand relationships of leaked data in underground communities.
- From March 2022 to March 2023, Breached users primarily requested technology, financial, and government sector data leaks. The financial and government sectors were the top victims of the data-leak advertisements resulting in a sale. The highest average prices by sector were aerospace at $1,458.42 USD, healthcare at $1,361.98 USD, and entertainment at $899.42 USD.
- During the same time period, Breached users primarily requested data leaks of victims located in the U.S., China, and Germany. The highest average prices by country were Germany at $2,251.47 USD, Colombia at $866.67 USD, and France at $573.06 USD. Of the data-leak advertisements resulting in a sale, the top victim countries were the U.S., Russia, and China.

BreachedForums Chronology

In April 2022, the U.S. Department of Justice (DOJ) and Europol announced their seizure of popular English-language eCrime forum RaidForums, which was founded in 2015.[1] Former RaidForums member *Pompompurin* used the seizure to promote their new forum, Breached.

By July 2022, Breached exceeded RaidForums' record count, with more than 11 billion records in its official data-leak section. Between August and December 2022, Breached maintained an average of 1,600 unique daily active users, surpassing RaidForums. Additionally, by September 2022, nearly 45 percent of the top 400 Breached monikers matched previous RaidForums monikers, indicating that eCrime actors almost certainly relocated from RaidForums to Breached.[2, 3]

[1] **CSDR-22069** CrowdStrike Intelligence Daily Report Day of 13 April 2022
[2] CrowdStrike Intelligence determined the top users through multiple factors, including length of time on the forum, quantity and location of posts and threads, number of thread replies, and interactions with other top users.
[3] **CSIT-22265** How the Threat Actor Pompompurin Quickly Grew Breached Following the Seizure of RaidForums

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In March 2023, the U.S. Federal Bureau of Investigation (FBI) arrested *Pompompurin*, and the Breached administrator *Baphomet* announced that the eCrime forum would be closing.[4] On 22 June 2023, the FBI seized the Breached clearnet domain and posted a seizure notice that included *Pompompurin*'s avatar in handcuffs.[5]

Advertisement Analysis Methodology

Despite the brief tenure of Breached, the forum's records offer insight into its members' behavior and trends in the greater eCrime ecosystem, as many members were almost certainly previous RaidForums members. Breached established itself as a preeminent data-leak marketplace by copying RaidForums' structure, adding six new data-leak subforums, and reducing the number of non-eCrime subforums.[6] eCrime actors often purchase leaked data from eCrime forums and then use the compromised credentials and personally identifiable information (PII) in phishing campaigns or identity-based attacks.

In August 2022, Breached subforums related to data-leaks maintained large numbers of active and unique users, as determined by the number of thread replies (Figure 1). The official Breached subforum maintained the highest average number of active users; the high level of user activity on other Breached subforums indicates that members were actively seeking data leaks.

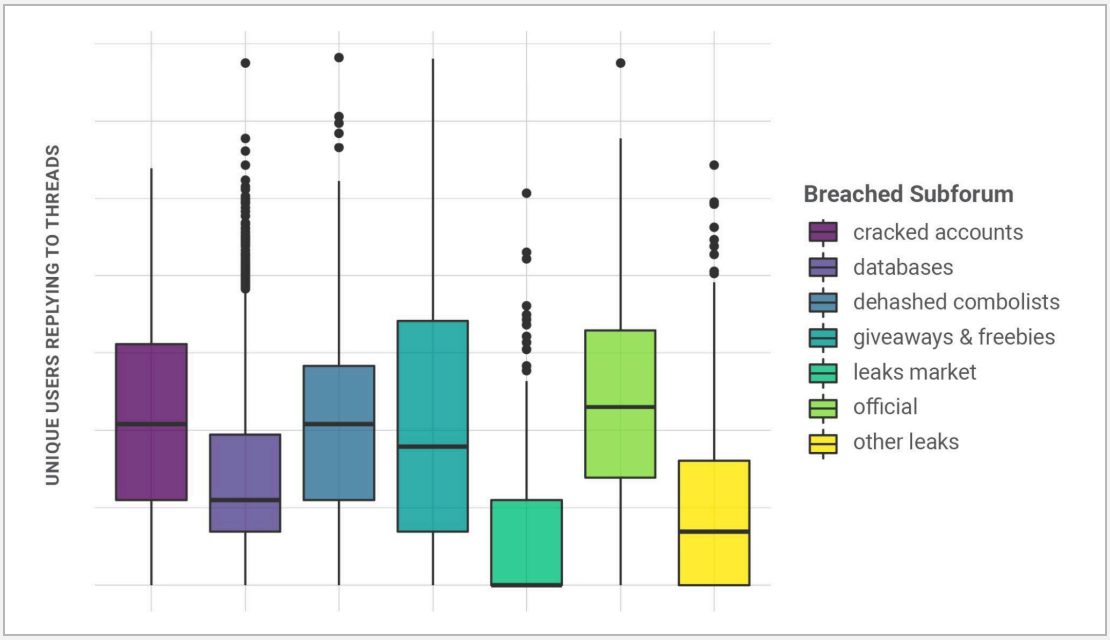

Figure 1. Active Users on Breached Data-Leak Subforums[7]

[4] **CSA-230448** FBI Arrests BreachForum Administrator Pompompurin; Forum Goes Offline
[5] **CSDR-23120** CrowdStrike Intelligence Daily Report 26 June 2023
[6] **CSIT-22265** How the Threat Actor Pompompurin Quickly Grew Breached Following the Seizure of RaidForums
[7] This diagram displays active users by Breached subforum. An active user is a forum member who creates a post. For each subforum in the diagram, the box's bottom line is the lower quartile of active users, the thick middle line is the median number of active users, and the top line is

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Analysis Methodology

The data-leak advertisements within Breached subforums offer numerous data points for assessing eCrime actor interest in victim data. Analyzing these advertisements shows the value of different types of data leaks to eCrime actors, specifically data-leak advertisers and purchasers.

To determine the value of data leaks on Breached, CrowdStrike Intelligence considered forum users' data-leak requests, average advertisement list price, and quantity of sold data leaks; these categories were analyzed by victim sector and country.[8] Limitations to the methodology include difficulty determining completed sales and thread replies serving as an imperfect indicator of demand. An additional limitation is imperfect price data, as the actual purchase price of completed sales may be negotiated via direct message and not shared publicly.

Sector-Level Findings

From March 2022 to March 2023, Breached users primarily requested technology, financial, and government sectors data leaks (Figure 2). These requests typically also specified a data type or database related to a particular country.

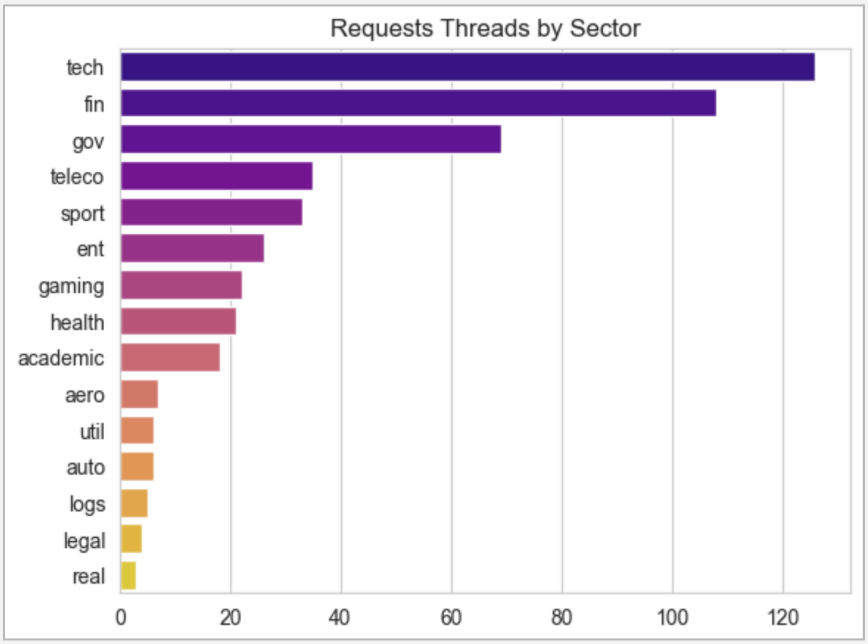

Figure 2. Most-Requested Sectors

the upper quartile of active users. The lines extending from each box represents the maximum number of active users, and the dots represent statistical outliers.

[8] This Tipper's *Appendix* lists all victim sectors, countries, and abbreviations.

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The sectors with the highest average data-leak list price were aerospace at $1,458.42 USD, healthcare at $1,361.98 USD, and entertainment at $899.42 USD (Figure 3).

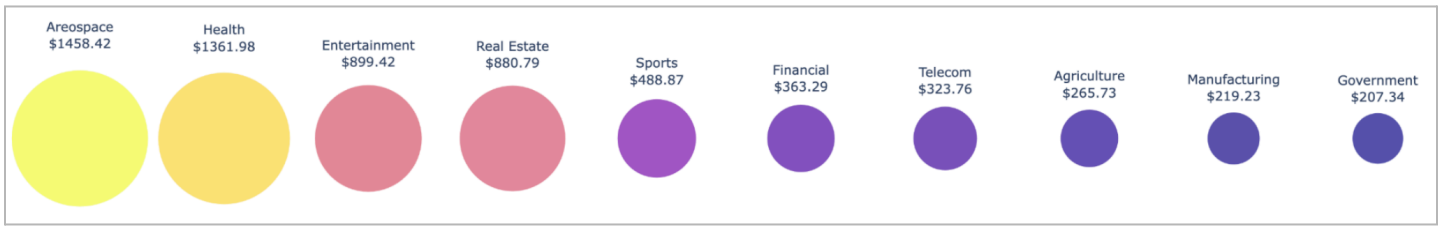

Figure 3. Average List Price by Sector

Calculating the Average List Price

The average list price of a data-leak advertisement reflects the typical advertiser's assessment of the data's worth; however, the data leak may ultimately sell for a different amount or not sell at all. Also, as averages are affected by outliers, the average list price may not be truly representative of data-leak advertisement pricing.

CrowdStrike Intelligence calculated the average list price by compiling all Breached advertisements across all subforums, including sold and unsold data leaks. Due to space constraints, this Tipper's average list price graphics include the top 10 entities (country or sector) rather than a complete listing of all entities and their average list price.

For the data leaks that sold (determined by the seller editing the advertisement and marking it as sold), the top sectors were financial and government (Figure 4).

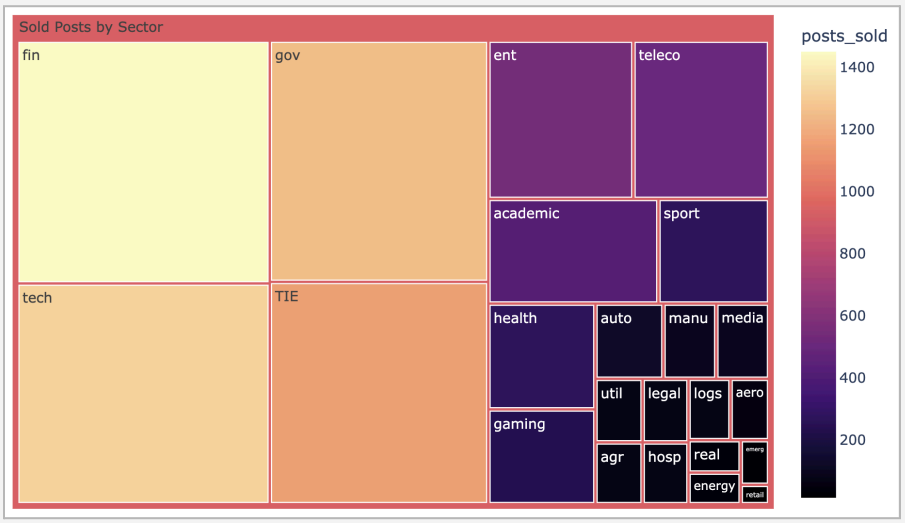

Figure 4. Sold Data Leaks by Sector

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These sector-level findings show that Breached users frequently request and sell financial and government sector data leaks. The average list prices for these sectors are not high relative to other sectors: $363.29 USD for financial, $207.34 USD for government, and $1,458.42 USD for aerospace.

Country-Level Findings

Breached users primarily requested data leaks of victims located in the U.S., China, and Germany (Figure 5). These requests typically also specified a certain type of data or database related to a specific country.

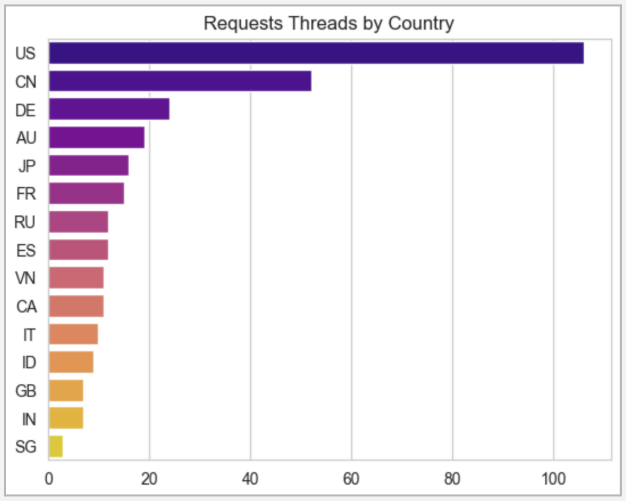

Figure 5. Most-Requested Countries

The countries with the highest average data-leak list price were Germany at $2,251.47 USD, Colombia at $866.67 USD, and France at $573.06 USD. The highest 10 list prices by country did not include the U.S. (Figure 6).

Figure 6. Average List Price by Country

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For the data leaks that sold, the top countries were the U.S., Russia, and China (Figure 7).

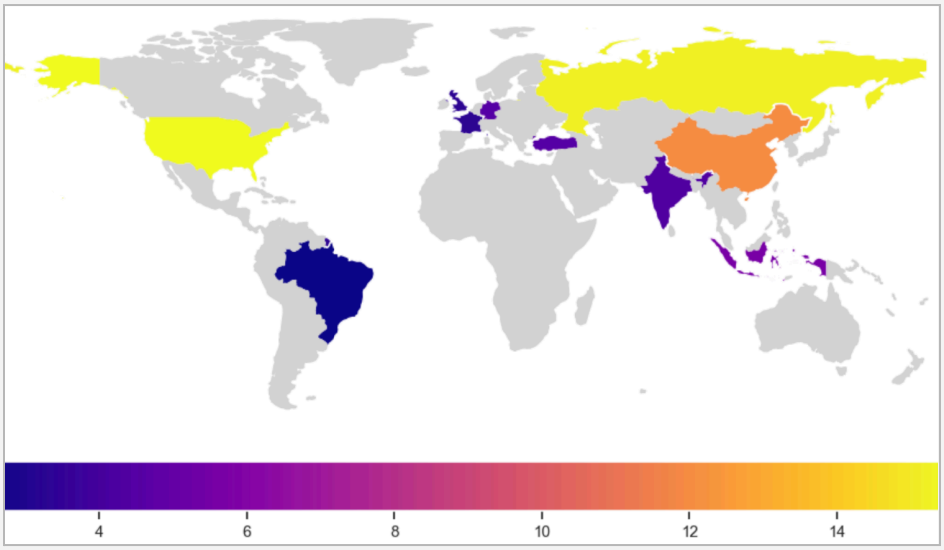

Figure 7. Percentage of Sold Data Leaks by Country

These country-level findings show that Breached users frequently request and sell the U.S. and China data leaks. Breached users also frequently request German data leaks, which have the highest average list price of all countries.

Conclusion

Following law enforcement's seizure of RaidForums, Breached established itself as a preeminent data-leak marketplace and maintained a high number of unique daily users and data-leak advertisements. Though law enforcement also seized Breached, the forum's former administrator *Baphomet* has tried to preserve the ethos of Breached by establishing BreachForums[.]vc—a new forum sharing a similar name and layout as BreachedForums.[9]

Baphomet's actions and eCrime actors' quick adoption of the new forum demonstrate the impact that an English-language data-leak forum has on the eCrime ecosystem. Moreover, analyzing Breached records shows the value of different data leak types, as demonstrated by eCrime actors who advertise and purchase compromised credentials.

[9] **CSA-230959** New BreachForums[.]vc Established by Likely Former BreachedForum Leadership; Large User Base Despite Early Data Leak Incident

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Appendix

Table 1 lists the Breached data-leak sectors with their corresponding abbreviation.

Academic (Academic)	Entertainment (Ent)	Logistics (Logs)	Technology (Tech)
Aerospace (Aero)	Finance (Fin)	Manufacturing (Manu)	Telecommunications (Teleco)
Agriculture (Agr)	Gaming (Gaming)	Media (Media)	Tourism (Hosp)
Automotive (Auto)	Government (Gov)	Real Estate (Real)	Utilities (Util)
Emergency (Emerg)	Health (Health)	Retail (Retail)	
Energy (Energy)	Legal (Legal)	Sports (Sport)	

Table 1. Sector List

Table 2 lists the Breached data-leak countries with their corresponding abbreviation.

Argentina (AR)	England (GB)	Israel (IL)	Peru (PE)	Spain (ES)
Australia (AU)	Finland (FI)	Italy (IT)	Peru (PE)	Sweden (SE)
Austria (AT)	France (FR)	Japan (JP)	Poland (PL)	Switzerland (CH)
Belgium (BE)	Germany (DE)	Kuwait (KW)	Portugal (PT)	Syria (SY)
Brazil (BR)	Greece (GR)	Lebanon (LB)	Qatar (QA)	Taiwan (TW)
Canada (CA)	Hong Kong (HK)	Luxembourg (LU)	Korea (KR)	Thailand (TH)
China (CN)	India (IN)	Malaysia (MY)	Romania (RO)	Turkey (TR)
Colombia (CO)	Indonesia (ID)	Mexico (MX)	Russia (RU)	Ukraine (UA)
Denmark (DK)	Iran (IR)	Netherlands (NL)	Saudi Arabia (SA)	United States (USA)
Egypt (EG)	Ireland (IE)	North Korea (KP)	Singapore (SG)	Vietnam (VN)

Table 2. Country List

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